

Problem

Find the maximum size of a subset of $\{1, 2, \dots, 50\}$ such that no two distinct elements sum to a multiple of 7.

Solution

Consider numbers modulo 7. Two numbers sum divisible by 7 if their remainders add to 7 (or both are 0).

- Remainder 0: pick at most 1 $\rightarrow$ 7 - Pairs of conflicting remainders (1, 6), (2, 5), (3, 4): pick the larger group.

Choose:

Remainder 0: 7

Remainder 1: 1, 8, 15, 22, 29, 36, 43, 50 (8 numbers)

Remainder 2: 2, 9, 16, 23, 30, 37, 44 (7 numbers)

Remainder 3: 3, 10, 17, 24, 31, 38, 45 (7 numbers)

Total: $1 + 8 + 7 + 7 = 23$

Example of a maximal subset:

$\{1, 2, 3, 7, 8, 9, 10, 15, 16, 17, 22, 23, 24, 29, 30, 31, 36, 37, 38, 43, 44, 45, 50\}$

Answer

23
